



BSR AutoMotor

Smart Home Water Motor Automation

User Manual & Feature Guide

The complete guide to installing, operating and getting the most out of your AutoMotor system — including the new built-in web control panel that runs in any phone or laptop browser.

For home owners, operators and installation technicians

Municipal Water Auto Start

Tank Auto Fill

Adjustable Tank Levels

5 Daily Schedules

Web Control Panel

Mobile App

Run-Time Statistics

Water Session Log

Remote Firmware Update

One-Tap Rollback

Data Export

Works Without Internet

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1. What Is the BSR AutoMotor System?

The BSR AutoMotor System is a smart, fully automated home water management solution. It exists to take away the daily chore of running your water motor by hand. Whether you are at home, at work or travelling, it watches your municipal (government / tap) water supply and your overhead or underground tank, and runs the pump automatically — exactly when it is needed, and never when it is not.

You set it up once, and the system takes care of the rest. It turns the motor **ON** when water is available or the tank is low, and **OFF** when the tank is full or the supply stops. On top of that you can see and control everything from your phone — either through the mobile app, or by simply opening the controller's own web page in a browser, with no app to install at all.

1.1 The Problem It Solves

Common problem	How BSR AutoMotor fixes it
Forgetting to turn ON the motor when municipal water arrives	The system detects water flow on its own and starts the motor within seconds
Forgetting to turn OFF the motor — the tank overflows	Tank level is monitored continuously; the motor stops the moment the tank is full
Motor running dry, or running when the tank is already full	Both the supply line <i>and</i> the tank level are checked before the motor is allowed to run
Water wasted through overflow	Automatic shut-off at the right level prevents overflow completely
No way to control the motor when away from home	Full remote control from the mobile app, from anywhere in the world
No way to know if water has arrived or the tank is full while you sleep	Distinct buzzer alerts at the motor <i>and</i> at the tank tell you exactly what happened
A fixed routine that cannot be adjusted	Up to five programmable schedules, plus thresholds you can tune yourself
Power cuts wiping out all settings	Every setting lives in permanent memory and is restored automatically
No record of how much water you actually received	Every supply session is measured in litres and logged, and can be downloaded as a file
No idea how long the motor has been running this month	Run time is totalled for every day of the last 31 days, split by what started the motor

1.2 System Components at a Glance

Component	Where it is installed	What it does
Host Controller	Near the motor / electrical panel	The brain of the system. Switches the motor, reads the flow sensor, keeps the clock, talks to the Sensor Node, serves the web control panel, and syncs with the cloud.
Sensor Node	Above or beside the water tank	Measures how full the tank is using a waterproof ultrasonic probe and reports the reading to the Host Controller every 5 seconds. Has its own buzzer.
Wi-Fi Extender	Between the controller and the tank	Repeats the Host Controller's own Wi-Fi network so the Sensor Node and your phone stay connected even when the tank is far from the motor.
Web Control Panel	Built into the Host Controller	A complete control page served by the controller itself. Opens in any browser on the same Wi-Fi. Nothing to install, and it works with no internet at all.
Mobile Application	Your Android smartphone	Live status, motor control, schedules and alerts — at home on local Wi-Fi, or from anywhere in the world over the internet.
Secure Cloud Service	The internet	Relays commands and status between your device and your phone when you are away from home, so the app keeps working on any connection.

1.3 What Is New in This Edition

If you are upgrading from an earlier AutoMotor release, everything you already know still works exactly as before. These capabilities have been added on top:

New capability	What it means for you
Built-in web control panel	The controller now serves its own control page. Open a browser on the same Wi-Fi and you get the full system — no app install, no internet, no account. Covered in detail in Section 5 .
Adjustable tank start / stop levels	The "tank is low" and "tank is full" marks are no longer fixed. Set them to suit your tank and your habits.
Adjustable water detection sensitivity	Tune how much flow counts as "municipal water has arrived", so a weak supply is still detected and a dripping line is not.
Motor run-time statistics	A daily bar chart for the last 31 days, with every day's run time split by what started the motor — schedule, municipal water, tank, app, web page or switch.
Municipal water session log	The last ten supply sessions, each with its start time, end time, duration and the litres collected.
Last run & next run	See at a glance what started the motor last, when, and when the next schedule is due to fire.
Restart history	A dated list of why the device last restarted — power on, brownout, update — with lifetime totals for each reason.
Remote silence for the tank buzzer	The buzzer on the roof can now be acknowledged from the control panel. It re-arms itself automatically, so it can never be left permanently disarmed.
Two firmware copies & one-tap rollback	An update is written into a spare slot, so the previous version stays in the device and can be restored instantly if anything looks wrong.
Local firmware upload	A technician can flash the device directly over Wi-Fi from a laptop with no internet connection at all.
Data download	Save the full run-time and water-session history to your phone as a plain text file.
Erase data & factory reset	Clear the history without touching your settings, or reset the whole device to as-new condition.

A note on connectivity. Internet is optional, not required. Every automation in this manual — municipal water auto start, tank auto fill, schedules, buzzer alerts and the entire web control panel — runs entirely inside the Host Controller. The internet connection only adds remote access from outside your home.

2. Host Controller — The Brain of the System

Installed at: near the water motor or the electrical panel.

Role: the Host Controller is the central unit. It reads the flow sensor, switches the motor, talks to the Sensor Node, keeps the clock, enforces every automation rule, manages your schedules, raises the alerts, serves the web control panel to your browser, and syncs with the cloud when internet is enabled.

2.1 Municipal Water Auto Start IMPROVED

Purpose: automatically starts the motor the moment municipal (tap) water arrives in your supply pipe — so you never miss a drop of incoming water.

In most homes municipal water does not arrive at a fixed time. This feature watches the supply pipe continuously with a flow sensor. The instant water starts moving, the system recognises it and turns the motor ON to pump that water into your tank.

- The flow sensor counts pulses from the pipe. Once the flow rate reaches the **detection threshold** — 0.5 litres per minute by default — the system confirms that municipal water has arrived.
- As a second check, a **minimum volume** must be collected within a two-minute rolling window (100 ml by default). This prevents a brief pressure surge or a dripping tap from being mistaken for a real supply.
- The motor starts automatically within seconds of detection. No action is needed from you.
- A **3-minute minimum run time** is enforced: the motor will not switch off for at least three minutes after starting. This stops false cut-outs caused by short fluctuations in flow.
- After that, the motor stops automatically when flow falls below the **cut-off threshold** (0.4 litres per minute by default), confirming the supply has ended.
- Every session records the **total quantity of water pumped in litres**, so you always know how much water each supply actually delivered.
- Both thresholds and the confirmation volume are **adjustable** — see section 2.7.
- Auto Start can be switched on or off at any time from the web panel or the mobile app.
- The setting is saved permanently and is restored automatically after a power cut.

Note. Auto Start will not fire if you have already turned the motor on yourself with Direct Motor Control. This prevents the motor being double-triggered.

Note. If the motor is already running under Direct Control when municipal water arrives, the system still sounds the buzzer so you know water has come — but it leaves the motor state alone.

2.2 Tank Auto Fill — Automatic Level Control

Purpose: keeps your tank topped up automatically by watching its level and running the motor when it drops — so you never run out of water at home.

In this mode the controller works together with the Sensor Node mounted above your tank. The node measures the level every five seconds and reports it. Based on that reading, the controller decides whether the motor should run.

- The motor turns **ON** automatically when the tank level falls to the **start level** or below — 50% by default.
- The motor turns **OFF** automatically when the level reaches the **stop level** or above — 87% by default.
- Both levels are adjustable from the web control panel (section 2.3).

The smart anti-chatter (hysteresis) system

Switching on and off at a single fixed level would make the motor flick on and off repeatedly as the water surface moves around that level. To prevent this, the system uses a layered decision instead:

Level reading	Decision	Reason
At or above the stop level	Tank full → motor OFF	The tank is full. There is nothing to pump.
Just below the stop level, when the tank was already marked full	Still treated as full → motor stays OFF	A small dead band that stops the motor restarting the instant it switched off.
Between the start and stop levels, motor not running	Treated as full → motor stays OFF	There is enough water. No action needed yet.
At or below the start level	Tank not full → motor ON	The tank is running low. Start refilling.
Below the stop level while the motor is already running	Tank not full → motor stays ON	Keep filling until it is genuinely full. Do not stop halfway.

Tip. This anti-chatter behaviour protects your motor from rapid repeated switching, which causes real wear and shortens motor life.

- If the Sensor Node stops reporting for more than **2 minutes** — for example a power cut at the tank — the controller treats the tank as full as a safety measure and stops the motor. A buzzer alert sounds immediately at the controller.
- The web panel also shows **how old the tank reading is**, so a stale number can never be mistaken for a live one.
- Tank Auto Fill is saved permanently and resumes automatically after a power cut.

Blocked. Tank Auto Fill cannot be switched on while the motor is already running under Municipal Water Auto Start or Direct Motor Control. This prevents two automations fighting over the same motor.

2.3 Adjustable Tank Start / Stop Levels NEW

Purpose: lets you decide how empty the tank must get before the motor starts, and how full it must be before the motor stops.

- The **start level** (default 50%) is the point at which the tank counts as low. Raise it if you would rather the tank was always kept near full; lower it if you prefer fewer, longer runs.
- The **stop level** (default 87%) is the point at which the tank counts as full.
- The two levels must be at least five percentage points apart, so the automation always has a sensible band to work in.
- Both are set from the **Tank Levels** card in the web control panel and are saved permanently in the device.

Why the stop level is capped at 87%. The ultrasonic probe sits above the full mark and, like all ultrasonic sensors, cannot measure anything closer than about 20–25 cm. Once the water rises past roughly 88%, the surface is inside the probe's blind zone and it stops returning a usable reading. A stop level set higher than that could never be reached — the motor would simply keep running until the tank overflowed. The system therefore refuses any value above 87%. See section 4.6 for how to raise this ceiling by remounting the probe higher.

2.4 Direct Motor Control

Purpose: gives you immediate manual control of the motor, overriding the automatic modes whenever you need to.

Sometimes you simply want the motor to run — to clean the tank, to test the installation, or to deal with something unusual. Direct Motor Control does exactly that, from the web panel or the mobile app.

- Turn the motor ON or OFF instantly. Works on local Wi-Fi and remotely over the internet.
- When you turn the motor ON manually, Tank Auto Fill is switched off automatically to avoid a conflict.
- When you turn the motor OFF manually, **all** automatic flags are cleared — Auto Start, Tank Auto Fill and municipal water tracking all reset cleanly.
- The motor state is saved permanently. If power fails while the motor is running, the motor resumes when power returns.
- A short buzzer beep confirms every manual ON and OFF.
- The system records that *you* started this run, so it shows up correctly in the run-time statistics and in the last-run record.

Blocked. A manual motor ON is refused if Municipal Water Auto Start has already detected water and started the motor, to prevent a double-trigger conflict. The panel tells you why.

2.5 Timer-Based Motor Scheduling

Purpose: runs the motor at a chosen time of day automatically — ideal where the municipal supply is predictable.

- Up to **five independent schedules** can be active at once.
- Each schedule sets a **start time** (hour, minute, second), a **duration** in minutes, and whether it **repeats daily** or runs just once.
- If you set a time that has already passed today, the system automatically moves it to the same time tomorrow.
- Repeating schedules roll forward by 24 hours after each run, so they keep going day after day with no further action from you.
- One-time schedules are cleared from memory automatically once they have run.
- All schedules survive a power cut.
- Schedules can be viewed, edited and deleted from the web panel and the mobile app.
- Deleting all schedules immediately stops any scheduled run that is in progress.
- The panel also shows you **when the next schedule is due** (section 5.10).

Blocked. Schedules will not run until the clock has been set. The panel shows a *"clock must be set first"* reminder in this case.

Blocked. A schedule will not take over the motor while Municipal Water Auto Start or Tank Auto Fill is actively running it. It is simply re-evaluated on the next cycle.

2.6 Flow Sensor & Water Quantity Measurement

Purpose: measures exactly how many litres of municipal water each supply session delivers — so you have a real picture of what you receive.

- A pulse-type water flow sensor is fitted to the pipe between the municipal supply and the motor.
- Each pulse represents a precisely calibrated volume of water.
- Pulses are counted continuously and converted into a live **flow rate** in litres per minute and a running **total volume**.
- The total is displayed live on the web panel and in the mobile app, and is pushed to the cloud when internet is enabled.
- The total resets to zero at the start of each new municipal supply session.

- Totals are written to permanent memory, so a restart mid-session does not lose the measurement.

2.7 Adjustable Municipal Water Detection NEW

Purpose: lets you tune how sensitive the system is to incoming water, so it suits your own supply pressure.

Every supply line behaves differently. A high-pressure connection surges; a weak one trickles. Three settings on the web panel let you match the system to yours:

Setting	Default	What it does
Flow rate to detect water	0.5 L/min	How much flow counts as "water has arrived". Lower it if a weak supply is being missed; raise it if the system reacts to a trickle that is not a real supply.
Flow rate to cut motor off	0.4 L/min	How little flow counts as "the supply has stopped". Should always be set below the detection value, so the system does not oscillate between the two.
Volume to confirm supply	100 ml	How much water must actually be collected within a two-minute window before the system accepts that a genuine supply is running.

Tip. Leave these at their defaults unless you have an actual problem. If the motor sometimes starts when no real water has arrived, raise the detection rate or the confirmation volume. If a genuine supply is being missed, lower the detection rate.

2.8 Municipal Water Session Log NEW

Purpose: keeps a record of your recent municipal water supplies, so you can see when water came and how much you got.

- The **last ten supply sessions** are stored in the device, newest first.
- Each entry shows the date and time water arrived, the time it stopped, how long the supply lasted, and the litres collected.
- A session that is still running is marked "**still flowing**" and its duration and volume update live.
- Very short bursts are deliberately ignored, so a pipe simply pressurising does not clutter the log with meaningless one-second entries.
- An in-progress session is checkpointed to permanent memory regularly, so a power cut costs at most a few minutes of the record rather than the whole session.
- The log can be downloaded as a text file (section 2.20).

2.9 Motor Run-Time Statistics NEW

Purpose: tells you exactly how long your motor has been running, day by day, and what caused each run.

- Run time is totalled for every day of the last **31 days** and kept in the device.
- Totals can be viewed for **today**, the **last 7 days** or the **last 30 days**.
- Every total is split by **what started the motor**: mobile app, web page, schedule, municipal water, tank auto-fill, or the physical switch.
- A daily bar chart shows the pattern at a glance; tapping any bar breaks that single day down by source.
- Days are split at your local midnight, so the figures line up with real calendar days.
- A run in progress is included in the totals as it happens.

Why this matters. Run time is the single best indicator of both your electricity use and your motor's wear. A month where "Tank auto-fill" suddenly dominates usually means a leak or a tap left open somewhere.

2.10 Last Run & Next Run Record NEW

Purpose: answers the two questions people actually ask — "why did the motor run?" and "when will it run next?"

- **Last started by** — which of the six sources started the most recent run.
- **Last started at** — the exact date and time it began.
- **Next schedule** — when the next stored schedule is due to fire, including repeating schedules rolled forward to tomorrow.
- All three survive power cuts.

2.11 Smart Buzzer Alert System

Purpose: keeps whoever is near the motor informed about important events, without needing to look at a phone.

The Host Controller has a built-in buzzer that uses different sound patterns for different events. This is genuinely useful when you are near the motor room and want to know what the system is doing without opening anything.

Pattern	Sound	Triggered by	Meaning
Continuous	One long steady beep	Municipal water arrived in Tank mode; Sensor Node disconnected; schedule completed; schedules deleted; device start-up	Something important happened. Check the panel or the app.
Fast beep	Rapid short beeps	Motor started by Tank Auto Fill	The motor is starting because the tank is low.
Beep-beep-pause	Two beeps, pause, repeat	Wi-Fi connection lost; weak Wi-Fi signal detected	Warning — a network or internet problem.
Medium alternating	Evenly paced on / off beeps	Municipal water detected by Auto Start; direct motor ON / OFF command	The motor has just been turned on or off.
Silent	—	System idle, everything normal	All is well. No action needed.

- Each alert has a maximum duration — around a second for simple confirmations, and up to a few minutes for a water-arrival alert that is meant to wake you.
- **Buzzer mute.** The controller's buzzer can be switched off entirely from the **Device Buzzer** tile on the web panel. While muted, no alert will sound. The setting is saved and survives a power cut, so remember to switch it back on.
- **Stop Buzzer** silences an alert that is sounding right now without disabling future alerts.
- Special case: if the motor is already running under Direct Control when municipal water arrives, the buzzer still sounds to notify you, even though the motor state does not change.

2.12 Silencing the Tank Buzzer from Anywhere NEW

Purpose: lets you acknowledge the buzzer on the roof from your phone, instead of climbing up to it.

The Sensor Node has its own buzzer, out at the tank, where nobody can conveniently reach it. It sounds whenever the tank level leaves the safe band. You can now silence it from the control panel:

- While the tank buzzer is actually sounding, a **red banner appears at the top of the web panel** carrying a single **Stop Buzzer** button.
- The same command is always available as **Stop Tank Buzzer** under **Controls**.
- The buzzer goes quiet within about five seconds of the tap.
- The tank buzzer and the controller's own buzzer are silenced separately, so quieting one never hides an alert from the other.

This is an acknowledgement, not a permanent setting. As soon as the tank level returns to the normal range, the silence is cleared automatically and the buzzer re-arms itself, ready for the next alarm. A mute that outlived its alarm would leave the roof buzzer disarmed for good after one press — and nobody would notice until the tank overflowed. As a second safeguard, the Sensor Node re-arms its own buzzer if the controller stops confirming the silence for ten minutes: an alarm nobody can hear is worse than one nobody can silence.

2.13 Built-in Real-Time Clock

Purpose: keeps accurate date and time inside the device so schedules always run at the right moment — even after a power cut.

- A dedicated hardware clock module with its own backup battery keeps time running while the mains power is off.
- The clock drives the schedule system and date-stamps every record the device keeps.
- The same module measures the ambient temperature to within a quarter of a degree, shown on the panel and in the app.
- When internet is enabled, the clock synchronises itself with global internet time servers automatically.
- You can also set it by hand from the web panel or the mobile app — useful on first setup or if it ever drifts.

2.14 Internet & Cloud Connectivity

Purpose: connects your system to the internet so you can monitor and control it from anywhere in the world.

When **Remote / IoT mode** is on, the Host Controller joins your home Wi-Fi router and communicates securely with a cloud service that relays commands and status between the device and your mobile app.

- Live status is uploaded every few seconds: motor state, tank level, water flow status, temperature, total water pumped and more.
- Commands from the app — motor on, motor off, set a schedule, change a mode — are delivered to the device and carried out in real time.
- Works whether your phone is on your home Wi-Fi or on mobile data anywhere in the world.

- All communication is authenticated. Only your own authorised account can send commands to your device.
- Remote mode can be turned on or off at any time. With it off, the system runs entirely on its own local Wi-Fi and every automation continues exactly as normal.

Privacy and security. All connection credentials are held securely inside the device and are never shown to users or exposed in the app or the web panel.

2.15 Smart Wi-Fi Management

Purpose: keeps the system connected and recovers from network problems on its own, with no manual intervention.

- The Host Controller always runs **its own local Wi-Fi network**, so the Sensor Node, the web panel and the mobile app have a working local connection even with no internet and no router.
- When Remote mode is on, the controller **also** joins your home router. Both connections run at the same time.
- Signal strength to the router is checked every few seconds. If it becomes too weak, a buzzer warning sounds and the system waits briefly for it to recover.
- If it does not recover, the controller disconnects cleanly from the router and falls back to local-only operation.
- If the router connection drops unexpectedly, reconnection is retried automatically for about a minute before falling back to local-only mode.
- When the router connection is restored, the system restarts itself to re-establish the cloud link and re-sync the clock.

Tip. Even with no internet at all, *every* automation keeps working — Auto Start, Tank Auto Fill, schedules, alerts, and the complete web control panel. Internet is only needed for remote access from outside your home.

2.16 Firmware Updates — Three Ways IMPROVED

Purpose: keeps your system current with new features and improvements, with a method to suit every situation.

Method	Needs	How it works
Cloud update (web panel or app)	Internet on the device	Tap Check For Update . The device asks the update service whether a newer version exists and answers yes or no. If yes, tap Update & Reboot — the device downloads, installs and restarts itself. Typically 1–3 minutes.
Local upload NEW	Only the local Wi-Fi	A technician selects a prepared firmware file on a phone or laptop and uploads it straight to the device over Wi-Fi, with a live progress bar. No internet needed anywhere.
Remote / cloud-triggered	Internet on the device	An update can be initiated through the cloud with no physical access to the device at all — useful for a service visit that does not need a visit.

During an update. Do not cut power to the device while an update is in progress. If an update is interrupted, the device falls back to its previous firmware copy automatically — see the next section.

2.17 Two Firmware Copies & One-Tap Rollback **NEW**

Purpose: makes updating safe. If a new version ever misbehaves, the previous one is still in the device and can be restored instantly.

- The device holds **two firmware copies**. An update is always written into the spare copy, so the version you were running stays intact.
- After an update the device runs the new copy. If it fails to come up and check in correctly, it **reverts to the previous copy automatically**.
- You can also roll back deliberately, at any time, with a single button on the web panel — no download and no internet required, because the older copy is already in the device.
- The panel shows which copy is running now and which is held in reserve, each with its version number and build date.
- Only one step back is kept. The next update overwrites the older copy.
- A factory reset does **not** erase either firmware copy, so rollback still works afterwards.

2.18 Permanent Settings Memory (Power-Cut Safe)

Purpose: nothing you configure is ever lost when the power goes off.

- Every setting is written to permanent memory inside the device the moment you change it.
- After a power cut the device restores: Municipal Water Auto Start on/off; the motor state (if it was running, it resumes); Tank Auto Fill on/off; all five schedules; tank start and stop levels;

water detection sensitivity; buzzer mute; Remote/IoT mode on/off; the water quantity counter; the water session log; run-time statistics; restart history; and the last-run record.

- You never need to reconfigure anything after an outage. The system picks up exactly where it left off.

2.19 Health Monitoring & Restart History IMPROVED

Purpose: records why the device restarted, so problems can be diagnosed quickly instead of guessed at.

- Every restart is recorded with its date, time and cause — normal power on, a software update, a power brownout (a voltage dip), a hardware fault, and others.
- The **last sixteen restarts** are kept, newest first, and shown on the web panel.
- **Lifetime totals** per cause are kept separately, so you can see, for example, "Power on × 9, Brownout × 2" even after the detailed list has rolled over.
- Health information is also uploaded to the cloud when internet is enabled, so the history can be reviewed from the app.
- Device uptime — how long since the last restart — is shown on the panel at all times.

What to look for. Repeated *Brownout* entries mean the supply voltage at the device is dipping, usually when the motor starts. That is worth showing to an electrician. Repeated *Power on* entries simply mean mains outages.

2.20 Data Export, Erase and Factory Reset NEW

Purpose: gives you control over the information the device has collected — to keep it, clear it, or start completely fresh.

Action	What it does	What it does <i>not</i> touch
Download Data	Saves the daily motor run time for every day still held in the device and all kept municipal water sessions as a plain text file on your phone or laptop.	Nothing is erased. Purely a copy.
Erase Stored Data	Deletes the water sessions, restart history, run-time statistics and last-run record, and restores detection sensitivity and the buzzer setting to defaults.	Your schedules and tank levels are kept.
Factory Reset	Erases <i>everything</i> the device has saved — schedules, tank and auto-start modes, water totals and all history. The device restarts as if brand new.	Both firmware copies are kept, so rollback still works.

Both erase actions ask you to confirm before anything is deleted, and neither can be undone. Download your data first if you want to keep it.

3. Physical Controls on the Host Unit

Besides the web panel and the mobile app, the Host Controller has physical inputs on the unit itself. This lets an operator standing at the motor act immediately, with no phone in hand.

3.1 Tank Mode Rocker Switch

Purpose: lets the operator at the motor enable or disable automatic tank filling without opening anything.

A rocker switch is mounted on the Host Controller unit. It directly controls whether Tank Auto Fill is active.

Switch position	State	Effect
Pressed / closed	Tank mode ON	Tank level monitoring starts. If the tank is at or below the start level, the motor turns on automatically.
Released / open	Tank mode OFF	Tank level monitoring stops. If the motor was running because of tank mode, it stops immediately.

- The switch is read every second, so a change takes effect at once.
- Its setting is saved to permanent memory and is remembered after a power cut.
- The state is mirrored on the web panel and in the app, so the switch and the software always agree.
- A run started this way is recorded as started by the **physical switch**, so it shows up correctly in the run-time statistics.

Blocked. The rocker switch is ignored while the motor is already running under Municipal Water Auto Start or Direct Motor Control. It takes effect only in neutral conditions.

3.2 Flow Sensor (Automatic Input)

Purpose: detects the presence and rate of water in the supply pipe — the core sensor behind Municipal Water Auto Start.

- The flow sensor is plumbed into the incoming water pipe and produces electrical pulses as water passes through it.
- The system counts these pulses every second to calculate flow rate and to decide whether municipal water is present.
- No manual action is ever needed — it works entirely automatically and feeds both the Auto Start logic and the litre measurement.

3.3 Temperature Sensor (Built In)

Purpose: provides an ambient temperature reading from inside the controller unit.

- The clock module includes a temperature sensor accurate to about a quarter of a degree.
- The reading is refreshed every second, shown live on the web panel and in the app, and uploaded to the cloud when internet is enabled.
- It is a useful rough indicator of how hot your electrical panel or motor room is getting.

4. Sensor Node — The Tank Level Monitor

Installed at: above or beside the water tank, overhead or underground.

Role: a compact, independent unit that measures the water level in your tank and reports it to the Host Controller every five seconds. Without the Sensor Node, Tank Auto Fill cannot work.

4.1 Ultrasonic Water Level Sensing

Purpose: measures the level inside your tank without ever touching the water, using sound.

The Sensor Node uses a waterproof ultrasonic probe mounted above the tank. It sends a sound pulse downward and times the echo coming back off the water surface. That time gives the distance to the water, which is converted into a level percentage.

- Tank depth and the probe's mounting height above the full mark are both configured to match your actual tank, and the offset is accounted for automatically.
- **Level % = (tank depth – distance down to the water surface) ÷ tank depth × 100**
- For a stable reading, **five measurements are taken every cycle and the middle (median) value is used**, which rejects noise and false reflections off the tank walls.
- If no usable echo comes back — a blocked or disconnected probe, or heavy foam — the reading is flagged rather than reported as though it were real, and the previous figure is clearly marked as held over.
- The web panel shows how old the reading is, so a stale value can never be mistaken for a live one.

Tip. The ultrasonic probe works with any water — clean, dirty or coloured — as long as the surface is in clear line of sight and nothing obstructs the space between the probe and the water.

4.2 Automatic Reporting to the Host Controller

Purpose: keeps the Host Controller continuously supplied with tank data so it can make real decisions about the motor.

- Every **5 seconds** the node measures the level and sends it to the Host Controller over the local Wi-Fi link.
- Each report carries the level percentage, the raw distance reading, whether the reading is fresh or held over, and whether the tank is currently in an alarm condition.
- If Wi-Fi drops, the node retries the connection by itself and resumes reporting as soon as it is back.
- The onboard LED blinks briefly on each successful report — a quick visual confirmation that the unit is alive and talking.

4.3 Local Tank Buzzer Alerts

Purpose: gives an audible warning right at the tank when the level is critically low or nearly full.

Condition	Buzzer	What it means
Level below 20%	Short beep every second	The tank is critically low. The motor should be running soon.
Level above 88%	Long beep every second	The tank is almost full. The motor should be stopping soon.
Level between 20% and 88%	Silent	Normal operating range. No alert needed.
Start-up complete	Two short beeps	The Sensor Node has booted and connected to the Host Controller.

Who decides the alarm. The alarm band belongs to the Sensor Node, because it is a property of where that particular probe is mounted. The Host Controller reads the node's verdict rather than working it out again from its own numbers — which means there is only ever one set of limits, and it cannot drift out of step.

4.4 Remote Buzzer Silence

Purpose: lets you acknowledge the tank buzzer from the control panel instead of going up to the tank.

Full details are in section 2.12. In short: while the tank buzzer is sounding, a red banner appears at the top of the web panel with a single **Stop Buzzer** button, and the same command is always available as **Stop Tank Buzzer** under Controls. The buzzer goes quiet within about five seconds, and re-arms itself automatically once the tank level returns to normal.

4.5 Wi-Fi Communication & Firmware Update

Purpose: connects the Sensor Node to the Host Controller reliably over the longest possible distance, and allows its software to be updated wirelessly.

- The node joins the Host Controller's own local Wi-Fi network automatically on start-up.
- The link is deliberately configured for **maximum range rather than maximum speed**: full transmit power, the longest-reaching Wi-Fi mode, and power-saving sleep disabled so no report is ever missed. The node is usually on a roof, far from the controller.
- If the connection drops it reconnects by itself, with no intervention.
- **LED indicator**: lit = Wi-Fi disconnected or error; off = connected and working; quick blink = a reading was just sent successfully.

- **Firmware update:** for the first **5 minutes** after every restart, the Sensor Node accepts a wireless software update from a browser on the same network. After five minutes that window closes automatically and the node devotes itself entirely to measuring and reporting.

Why the update window closes. Keeping an update service running permanently on a device nobody can physically reach would be a standing risk. A short window that opens only when someone deliberately restarts the node gives easy maintenance without leaving that door open.

4.6 Mounting the Probe Correctly

Purpose: getting this right is the single biggest factor in how accurate your tank readings will be.

Every ultrasonic probe has a **blind zone** — a minimum distance below which it cannot measure at all. On the probe used here that is roughly 20–25 cm. If the probe is mounted only a little above the full mark, the water surface enters that blind zone while the tank is still filling, and the top of the tank is effectively unmeasurable.

Water level	Distance the probe sees	Result
100% (full)	≈ 12 cm	Inside the blind zone — no reading
94%	≈ 17 cm	Inside the blind zone — no reading
88%	≈ 22 cm	Marginal
20%	≈ 78 cm	Reliable

Recommendation for installers. Mount the probe at least **25 cm above the tank's full mark** and set the mounting offset to match. Done that way, the whole usable range of the tank is genuinely measured and the stop level can be set higher with confidence. With a shorter offset, treat "full" as an inference rather than a measurement — which is precisely why the system caps the stop level at 87% by default.

5. The Web Control Panel

The Host Controller serves its own complete control page. Open a browser on any phone, tablet or laptop connected to the system's Wi-Fi and the whole system is in front of you — live status, motor control, schedules, settings, history and firmware management. **There is nothing to install, no account to create and no internet connection required.** The screenshots in this section are of the actual panel.

5.1 Opening the Panel

1. On your phone or laptop, open Wi-Fi settings and connect to the Host Controller's own network, named **ESP32-WROOM**. The password is supplied with your installation.
2. Open any web browser and go to **http://192.168.4.1**
3. The panel loads and begins showing live readings immediately.
 - The panel is designed phone-first: the motor state, the power button and the mode tiles all fit on one screen on a typical phone, with no scrolling needed for day-to-day use.
 - Everything below that is grouped into collapsible sections. Tap a section heading to open or close it, or use **Expand all** to open everything at once.
 - The header carries a connection indicator — a green dot while the device is responding, red if it is not.
 - The first time you open the panel it asks you to tap once to enable the municipal-water alert sound. Browsers require that one tap before any page is allowed to play audio.

Tip. Add the page to your phone's home screen. It then opens like an app, full-screen, straight to the motor controls.

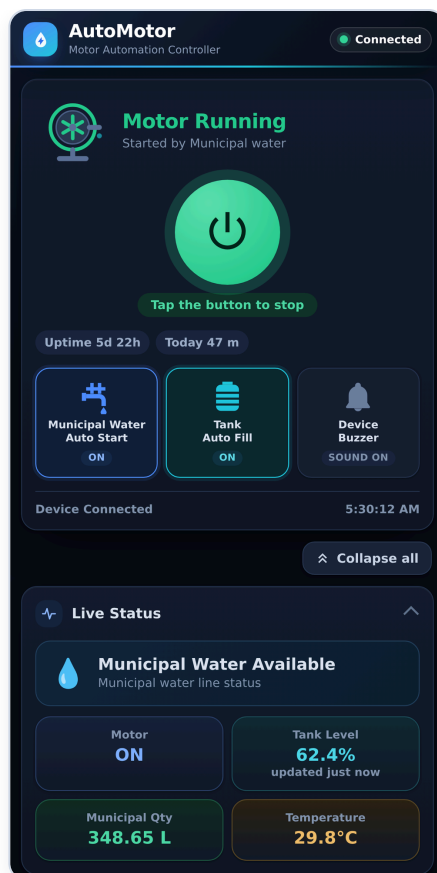


Figure 5.1 — The panel as it opens: connection state in the header, the motor card, and the collapsible sections below.

5.2 The Motor Card

Purpose: the one screen that answers "what is my motor doing, and how do I change it".



Figure 5.2 — The motor card. Green means the motor is running; the subtitle names what started it.

Element	What it shows / does
Animated motor graphic	The impeller spins while the motor is running, so the state is obvious at a glance even across a room.
Motor Running / Motor Off	The current state, with a subtitle naming what started this run — for example "Started by Municipal water".
Power button	One tap starts or stops the motor. You are asked to confirm before anything switches, so a mis-tap cannot start your pump.
Uptime	How long since the device last restarted.
Today	How long the motor has run so far today, across all sources.
Municipal Water Auto Start tile	Tap to turn automatic starting on incoming water on or off. Shows ON or OFF.
Tank Auto Fill tile	Tap to turn automatic tank level control on or off. Shows ON or OFF.
Device Buzzer tile	Tap to mute or un-mute the controller's own buzzer. Shows SOUND ON or MUTED.
Footer line	Connection status on the left, the time of the last successful reading on the right.

If a request is refused. When a mode conflict blocks an action — for example turning the motor on manually while Auto Start is already running it — the panel says so in plain language directly under the button, rather than silently doing nothing. The full set of rules is in section 8.2.

5.3 Live Status

Purpose: the complete real-time picture of your water system in one place.

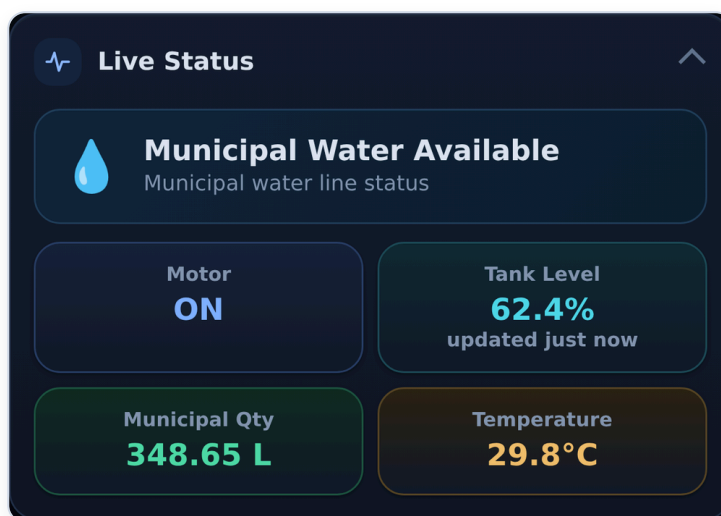


Figure 5.3 — Live Status. The banner reports the municipal supply; the four tiles carry the key readings.

Reading	Shows	Why it matters
Municipal water banner	Available / not available	Whether government water is flowing in your pipe right now.
Motor	ON or OFF	Whether your motor is running at this moment.
Tank Level	Percentage, to one decimal place, with the age of the reading	Exactly how full your tank is — and how fresh that figure is. "Updated just now" means the Sensor Node is reporting normally.
Municipal Qty	Litres in the current session	How much water this supply has delivered so far.
Temperature	°C at the controller	How hot the motor panel environment is getting.

Tip. If the tank level says "no reading yet" or shows an age of several minutes, the Sensor Node is not reaching the controller. Start with section 10, *Troubleshooting*.

5.4 Controls

Purpose: the occasional actions that do not belong on the main motor card.

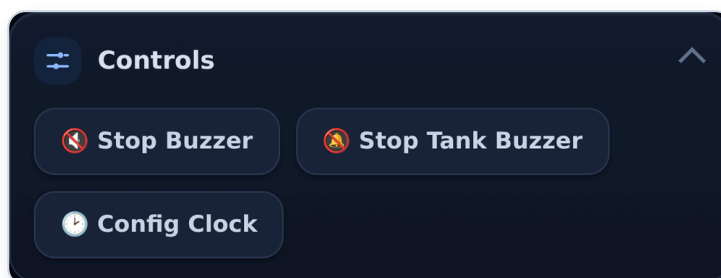


Figure 5.4 — Controls.

Button	What it does
Stop Buzzer	Silences an alert sounding at the controller right now. Future alerts still sound.
Stop Tank Buzzer	Silences the buzzer out at the Sensor Node. Takes effect within about five seconds and re-arms automatically once the tank level is normal again.
Config Clock	Opens a date-and-time box so you can set the device clock by hand. Needed before schedules will run, unless the device has already synced from the internet.

5.5 Schedules

Purpose: set, review and clear the five daily motor schedules.

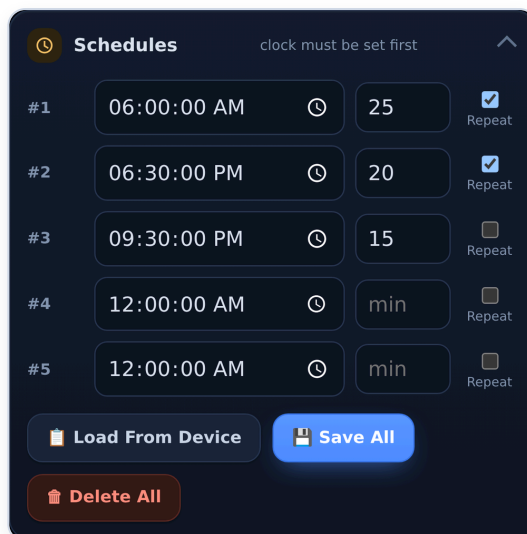


Figure 5.5 — Schedules. Two repeating daily runs and one that fires only once.

Field / button	What it does
Slots #1 to #5	Five independent schedules. Leave a slot blank to disable it.
Time	The start time, to the second.
min	How many minutes the motor should run for.
Repeat	Ticked = runs every day at that time. Unticked = runs once, then clears itself.
Load From Device	Fetches the schedules currently stored in the controller, so you can see and edit exactly what is really set.
Save All	Writes all five slots to the device permanently.
Delete All	Clears every schedule. Any scheduled run in progress stops at once. You are asked to confirm first.

Set the clock first. The section header carries the reminder "clock must be set first" for good reason — schedules are refused until the device knows the real time, because an unset clock would run your motor at the wrong hour.

5.6 Remote / IoT Mode

Purpose: one switch that controls whether the device connects to the internet for remote access.

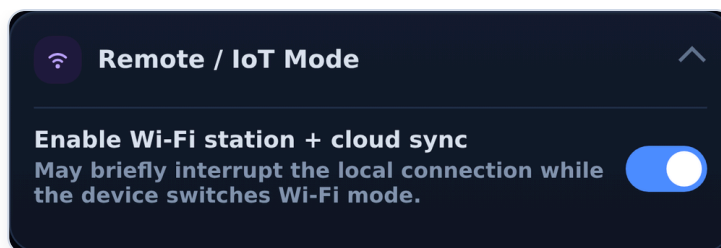


Figure 5.6 — Remote / IoT mode.

Position	Effect
On	The controller joins your home router, syncs its clock from the internet, connects to the cloud service, and remote control from the mobile app becomes available anywhere in the world.
Off	The controller disconnects from the router and runs on its own local Wi-Fi only. All automation continues unchanged. Remote app control is unavailable, but local control on the same Wi-Fi keeps working normally.

Expect a brief interruption. Switching this setting makes the device change Wi-Fi mode, so the panel may lose contact for a few seconds and then reconnect by itself. That is normal.

5.7 Firmware — Cloud Update

Purpose: updates the device over the internet, in two taps.

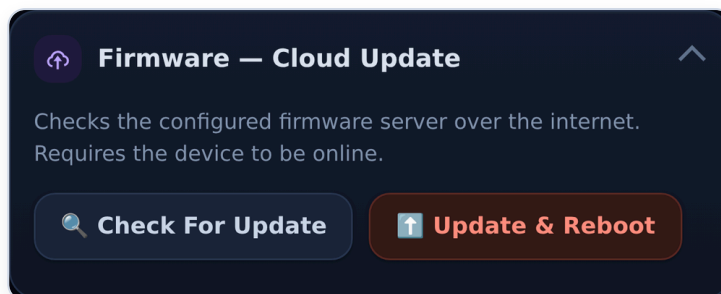


Figure 5.7 — Cloud update.

1. Tap **Check For Update**. The device asks the update service whether a newer version exists and reports back yes or no.
2. If a newer version is available, tap **Update & Reboot**. You are asked to confirm.
3. The device downloads, installs and restarts itself. This normally takes one to three minutes, after which the panel reconnects on its own.

Requires internet. Remote / IoT mode must be on and the device online. If it is not, use the local upload method instead.

5.8 Firmware — Local Upload

Purpose: lets a technician update the device with no internet connection at all.

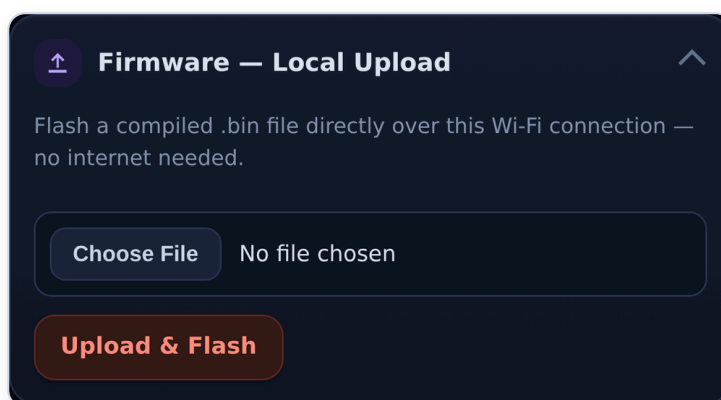


Figure 5.8 — Local upload, with its progress bar.

- Select a prepared firmware file supplied by BSR, then tap **Upload & Flash**.
- The file transfers straight over the local Wi-Fi link, with a live progress bar.
- The device installs it and restarts automatically.
- Nothing leaves your premises and no internet connection is used at any point.

For service use. Only ever upload a firmware file supplied by BSR for this product. Do not interrupt the power while the progress bar is moving. If an upload does fail, the previous firmware copy is still in the device — see section 5.15.

5.9 Motor Run Time

Purpose: shows how long the motor has actually run, day by day, and what caused each run.



Figure 5.9 — Motor Run Time over 30 days, split by what started each run.

Element	What it shows
Today / 7 days / 30 days	Choose the period. Everything below updates to match.
Total run time	The combined motor run time for the selected period.
Source breakdown	The same total split six ways — Mobile app, Web page, Schedule, Municipal water, Tank auto-fill and Physical switch.
Daily bar chart	One bar per day, oldest at the left, dated along the bottom. Tap any bar to see that single day broken down by source.

- The device keeps up to **31 days** of daily history.
- Days are split at your local midnight, so the bars line up with real calendar dates.
- A run that is in progress right now is already counted in the totals.

Reading the chart. Steady bars with municipal water dominating is a healthy pattern. A sudden climb in "Tank auto-fill" means the tank is emptying faster than usual — a running toilet, a dripping tap or a leak. A climb in "Schedule" usually means a schedule duration is set longer than it needs to be.

5.10 Last & Next Run

Purpose: answers "why did the motor run?" and "when will it run next?" without any digging.

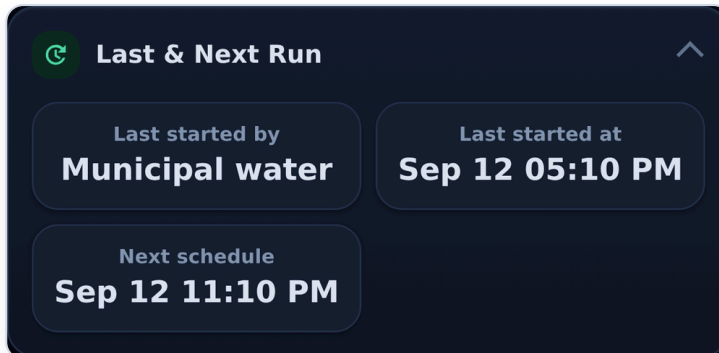


Figure 5.10 — Last & Next Run.

- **Last started by** — which of the six sources started the most recent run.
- **Last started at** — the date and time it began.
- **Next schedule** — when the next stored schedule will fire, with repeating schedules already rolled forward.

5.11 Municipal Water — Last 10 Sessions

Purpose: the record of when municipal water actually came, and how much of it you got.



Figure 5.11 — The ten most recent supply sessions, newest first. The top entry is still in progress.

- Each row shows the date and time water arrived, the litres collected, when it stopped, and how long the supply lasted.

- A session still in progress is marked **"still flowing"**, and its duration and litres keep climbing while you watch.
- Scroll within the list to see all ten.

Tip. Over a few weeks this list tells you the real pattern of your municipal supply — what time it usually arrives and roughly how much it delivers. That is the information you need to set a sensible schedule.

5.12 Reset History

Purpose: shows why the device has restarted, which is where most diagnosis of an intermittent problem begins.

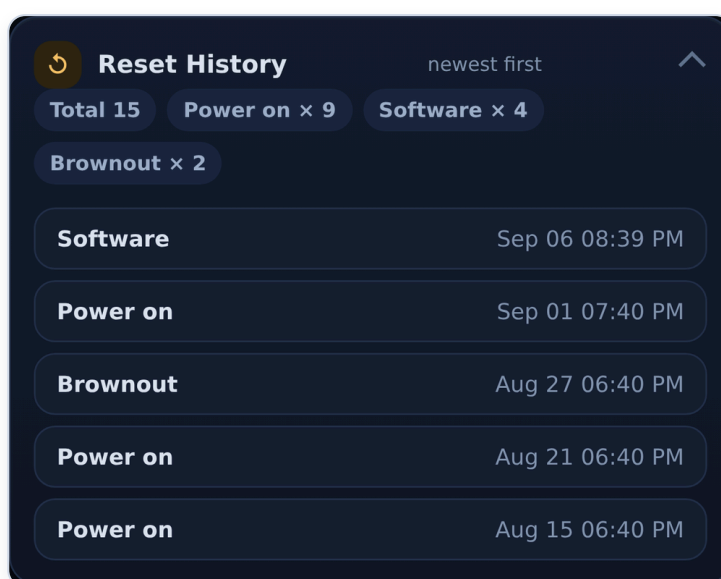


Figure 5.12 — Reset History, with lifetime totals across the top.

Reason shown	What it means
Power on	Mains power was removed and restored. A normal power cut.
Software	The device restarted itself deliberately — most often after an update or a settings change that requires a restart.
Brownout	The supply voltage dipped too low for the device to keep running. Commonly caused by the motor starting on a weak supply.
Watchdog / Panic	A fault was detected and the device recovered itself automatically. Isolated entries are normal; repeated ones are worth reporting.
External	A hardware reset, for example the reset button being pressed.

- The **last sixteen** restarts are listed, newest first, each dated and timed.
- The pills across the top are **lifetime totals** for each cause, which keep counting long after individual entries have scrolled off the list.

5.13 Tank Levels

Purpose: sets the two levels that drive Tank Auto Fill.

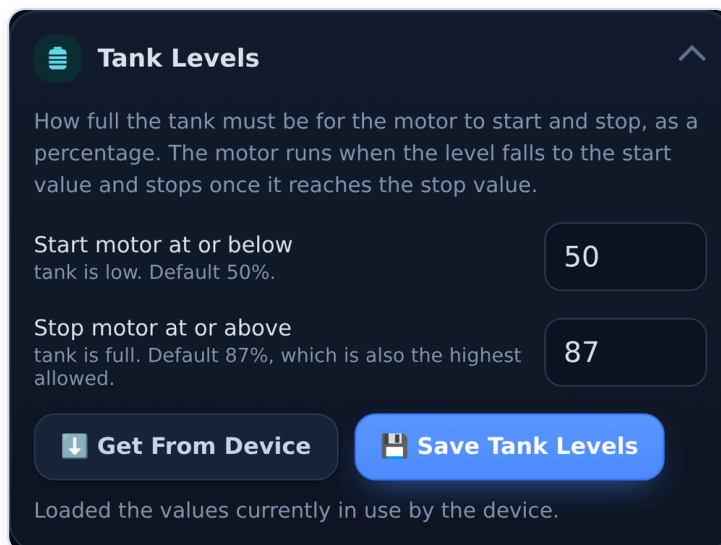


Figure 5.13 — Tank Levels.

Setting	Default	Meaning
Start motor at or below	50%	The tank counts as low at this level, and the motor starts.
Stop motor at or above	87%	The tank counts as full at this level, and the motor stops. 87% is also the highest value the system will accept.

- **Get From Device** loads the values currently in use, so you always edit from the truth rather than from a guess.
- **Save Tank Levels** writes them permanently.
- The two levels must be at least five percentage points apart.

If you try to set the stop level above 87%, the panel explains why it cannot: above roughly 88% the water surface is inside the ultrasonic probe's blind zone, so the motor would never see the tank reach that figure and would keep running until the tank overflowed. Section 4.6 explains how to raise this ceiling properly, by remounting the probe higher.

5.14 Municipal Water Detection

Purpose: tunes how sensitive the system is to incoming water.

Figure 5.14 — Detection sensitivity.

- **Flow rate to detect water** (default 0.5 L/min) — how much flow counts as water having arrived. Lower is more sensitive.
- **Flow rate to cut motor off** (default 0.4 L/min) — how little flow counts as the supply having stopped. Keep this below the detection value.
- **Volume to confirm supply** (default 100 ml) — how much water must actually be collected in a two-minute window before a supply is accepted as genuine.
- **Get From Device** loads the values in use; **Save Sensitivity** stores your changes permanently.

Adjust only if you have a symptom. Motor starting when no real water has arrived → raise the detection rate or the confirmation volume. A genuine weak supply being missed → lower the detection rate.

5.15 Firmware Slots & Rollback

Purpose: shows which firmware is running, which is held in reserve, and lets you go back one step instantly.



Figure 5.15 — Firmware slots. The running copy on the left, the previous copy held in reserve on the right.

- **Running now** — the firmware the device is using at this moment, with its version number and the date it was built.
- **Previous / spare** — the copy the last update replaced, still intact in the device.
- **Roll Back to Previous Firmware** — restores that older copy on the next restart. No download and no internet are needed, because the copy is already there. You are asked to confirm first.
- The button is disabled when there is no valid previous copy to go back to.
- Only one step back is kept. The next update overwrites the older copy.

Automatic safety net. Immediately after an update, the device must start up and confirm it is healthy. If it cannot, it reverts to the previous copy by itself — you do not have to be there, and you do not have to notice.

5.16 Maintenance

Purpose: download, clear or completely reset the information the device has stored.

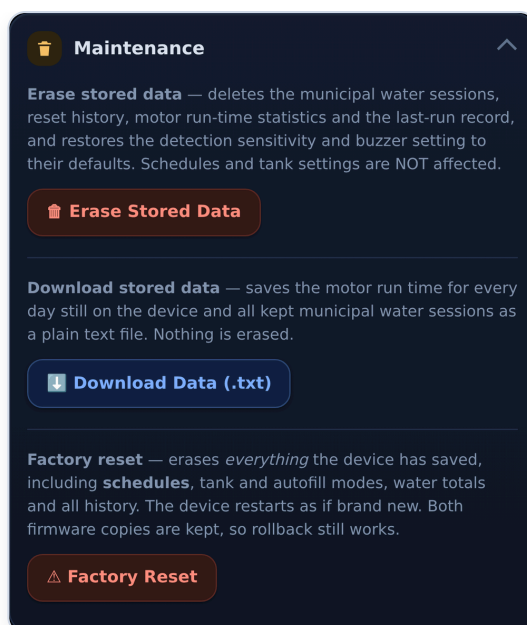


Figure 5.16 — Maintenance. Each action states exactly what it affects.

Action	Effect
Erase Stored Data	Deletes water sessions, reset history, run-time statistics and the last-run record, and restores detection sensitivity and the buzzer setting to defaults. Schedules and tank levels are kept.
Download Data (.txt)	Saves the daily run time for every day still held, plus all kept water sessions, as a plain text file on your phone or laptop. Nothing is erased.
Factory Reset	Erases everything saved — schedules, modes, water totals and all history — and restarts the device as if brand new. Both firmware copies are kept , so rollback still works afterwards.

Both erase actions ask for confirmation and cannot be undone. Download your data first if you want to keep it.

5.17 Diagnostics

Purpose: a running log of what the panel and the device are saying to each other — useful when reporting a problem to support.

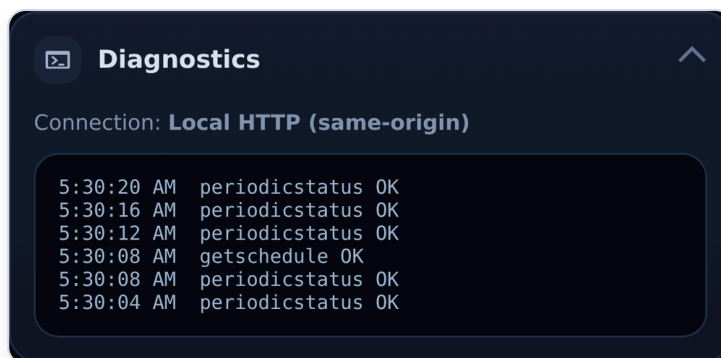


Figure 5.17 — Diagnostics.

- Shows how the panel is connected to the device.
- Keeps a timestamped log of recent exchanges, newest at the top.
- You never need this for normal use — but if you contact support, a photograph of this panel is often the fastest way to get to the answer.

5.18 The Tank Buzzer Alarm Banner

Purpose: tells you the roof buzzer is sounding, and offers the one thing anybody wants from a screaming buzzer.

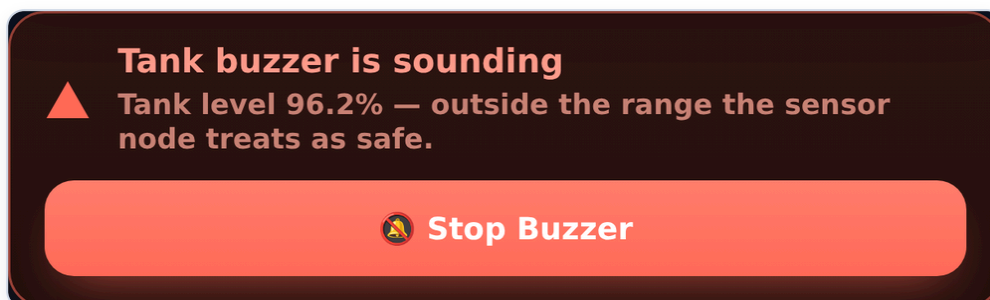


Figure 5.18 — The alarm banner. It appears only while the Sensor Node's buzzer is actually sounding.

- The banner appears at the very top of the panel, above everything else.
- It states the current tank level and that it is outside the range the Sensor Node treats as safe.
- It carries exactly one button: **Stop Buzzer**.
- It disappears by itself once the tank level returns to the normal range — and the buzzer re-arms at the same moment, ready for the next alarm.

6. Mobile Application

The BSR AutoMotor app for Android gives you full visibility and control of your system — at home on local Wi-Fi, or from anywhere in the world over mobile data. It covers the same core functions as the web panel, plus the remote access that only the cloud connection can provide.

Web panel or app — which should I use? Use the **web panel** when you are at home: it needs no installation, works with no internet, and carries the deeper settings and history. Use the **app** when you are away from home, or when you want push notifications. Both talk to the same device and both show the same truth.

6.1 Live Dashboard

Reading	Shows	Why it matters
Motor status	ON / OFF	Know instantly whether the motor is running.
Municipal water status	Flowing / no water	See whether government water is in your pipe.
Water tank level	0% to 100%	Know how full the tank is without going to look.
Tank Auto Fill mode	ON / OFF	See whether automatic filling is active.
Auto Start status	ON / OFF	See whether automatic start on water arrival is active.
Total water pumped	Litres, current session	Track how much the latest supply delivered.
Temperature	°C at the controller	Monitor the motor panel environment.
Sensor Node status	Reading from the tank unit	Confirm the tank unit is alive and reporting.

6.2 Motor Control

Button	Action	Possible response
Motor ON	Turns the motor on immediately (Direct Control)	Starts — or is blocked, with a message, if Auto Start has already started the motor on incoming municipal water.
Motor OFF	Turns the motor off and clears all automatic flags	Stops. Every automatic mode is reset cleanly.
Motor Status	Asks the device for its current state	ON or OFF confirmation.

6.3 Mode Controls

Control	Effect
Water Tank Mode ON	Enables automatic tank level control. Blocked if the motor is already running under Auto Start or Direct Control.
Water Tank Mode OFF	Disables it. Works at any time; the motor stops if it was running because of tank mode.
Start AutoFill	The system will now run the motor automatically every time municipal water is detected.
Stop AutoFill	Water is still detected and reported, but the motor is not started automatically — useful when you want to skip one particular supply.

6.4 Other App Functions

- **Stop Buzzer** — silences the controller's buzzer from your phone. It resumes automatically if the condition that raised it still exists.
- **Schedules** — set a start time, duration and daily repeat; view all stored schedules; delete them all with one tap. Up to five.
- **Set Clock** — sets the device's date and time from your phone. Not needed when Remote mode is on, since the clock syncs from the internet automatically.
- **Internet (IoT) toggle** — turns remote access on or off, exactly as the web panel does.
- **Device firmware update** — check for a newer version and install it, without touching the device.
- **App self-update** — the app checks for a newer version of itself and prompts you to install it, so you always have the latest features without visiting a store.
- **Ping** — a quick check that the device is alive and responding.
- **Water status check** — confirms whether municipal water is being detected right now.

Clock reminder. If the clock has not been set, the app will show *"Please set the clock"* and refuse to accept schedules. Set it once and the hardware clock keeps it, even through power cuts.

7. Wi-Fi Extender — Bridging the Distance

Installed at: roughly midway between the Host Controller at the motor and the Sensor Node at the tank.

Role: repeats the Host Controller's own Wi-Fi network so the Sensor Node stays connected even when it is installed a long way away.

7.1 Range Extension

Purpose: solves the single most common installation problem — the tank being too far from the motor for a direct Wi-Fi link.

- The Host Controller's local Wi-Fi network covers a limited range, typically 10–20 metres inside a building.
- If your tank is on a rooftop while the motor is on the ground floor, the direct signal may be too weak to be reliable.
- A standard household Wi-Fi range extender, placed at the midpoint, bridges the gap.
- It repeats the controller's network under the same name, and the Sensor Node connects through it without knowing the difference. No configuration is needed in either device.

Tip. Any ordinary Wi-Fi repeater will do, configured to repeat the Host Controller's network (**ESP32-WROOM**). No special hardware is required. Place it where it still has a strong signal from the controller — a repeater in a weak spot simply repeats a weak signal.

How to tell whether you need one. Open the web panel and look at the tank level on the **Live Status** card. If it regularly says "updated 2 m ago" or worse, or drops to "no reading yet", the link is marginal and an extender will fix it.

8. Which Mode to Use & When

The system has four ways to start the motor. Understanding when each one earns its place is how you get the most out of the installation.

8.1 Choosing a Mode

Your situation	Recommended	How to enable it
Municipal water arrives at unpredictable times and you want the motor to run every time it does	Municipal Water Auto Start	Leave the Auto Start tile ON — it is the default. Nothing else to do.
You want the tank simply kept full, regardless of when water comes	Tank Auto Fill	Tap the Tank Auto Fill tile ON, or flip the rocker switch on the unit.
You need the motor running right now — tank cleaning, testing, an emergency	Direct Motor Control	Tap the power button on the motor card. Tap it again when you are done.
Municipal water arrives at the same time every day	Schedules	Set a repeating schedule with the start time and duration.
You are going away and want everything handled without you	Auto Start + Tank Auto Fill together	Turn both tiles ON. Between them they cover every scenario.

8.2 Mode Priority & Conflict Rules

The system has built-in rules so that two modes can never fight over the same motor. Every refusal comes with a plain-language explanation on screen.

Situation	What happens	Why
Auto Start has already started the motor on incoming water — you tap Motor ON	Blocked , with a message	The motor is already running because of Auto Start. Starting it again would double-trigger it.
Motor is on via Auto Start — you try to enable Tank Auto Fill	Blocked	Tank control cannot take over while a municipal water session is running the motor.
Motor is on via Direct Control — municipal water arrives	Motor stays on ; buzzer sounds for a few minutes	No conflict — the motor is already doing the right thing. The alert simply tells you water has also come.
Motor is on via Direct Control — you try to enable Tank Auto Fill	Blocked	Automatic control should not take over while you have deliberately taken manual control.
A schedule is due — but Auto Start or Tank Auto Fill is already running the motor	The schedule does not activate	The live session takes priority. The schedule is simply re-evaluated on the next cycle.
Tank Auto Fill is running — the Sensor Node stops reporting for 2 minutes	Motor OFF ; buzzer alert; tank assumed full	A safety measure. Never run the motor without tank data — it could overflow the tank.
You set a schedule but the clock has never been set	Schedule refused , with a "set the clock" message	Schedules depend on real time. An unset clock would run the motor at the wrong hour.
You turn the motor OFF manually	Everything clears	Manual off is absolute: Auto Start, Tank Auto Fill and municipal tracking all reset cleanly, so nothing restarts behind your back.

8.3 Three Worked Examples

A. A typical city home with an unpredictable supply

- Municipal Water Auto Start: **ON** — catch every supply automatically.
- Tank Auto Fill: **ON**, start 50%, stop 87% — top up from the underground sump on the days no municipal water comes.
- Schedules: none needed.
- Check the **Municipal Water sessions** list after a fortnight to learn your real supply pattern.

B. A supply that arrives at the same time every day

- Watch the sessions list for a week to find the real arrival time and typical duration.
- Set schedule #1 to that time, with a duration slightly longer than a typical session, and **Repeat** ticked.
- Keep Auto Start **ON** as well — it will catch any day the supply comes early or late.

- Use **Last & Next Run** to confirm the next fire time is what you expected.

C. Away from home for two weeks

- Turn Remote / IoT mode **ON** before you leave, and confirm the app connects from mobile data while you are still at home.
- Auto Start **ON** and Tank Auto Fill **ON**.
- Consider muting the Device Buzzer if the unit is near a neighbour — but remember to unmute it when you return.
- On your return, check **Motor Run Time** over 30 days. An unusual amount of "Tank auto-fill" while the house was empty points to a leak.

9. Alert Reference Guide

Every alert the system can produce, what it means and what — if anything — you should do about it.

9.1 Host Controller Buzzer

Sound	What it means	What to do
One continuous beep at start-up	The device has powered on and started successfully.	Nothing. Normal.
Long continuous beep during normal running	Municipal water arrived while in tank mode, a schedule finished, schedules were deleted, or the Sensor Node has gone quiet.	Open the panel or the app to see which. If it is the Sensor Node, see section 10.
Rapid short beeps	The motor has just started because the tank is low.	Nothing — this is Tank Auto Fill doing its job.
Two beeps, pause, repeating	Wi-Fi lost, or the signal to your router has become weak.	Check your router. The system falls back to local-only operation by itself and keeps automating.
Evenly paced on / off beeps	Municipal water was detected, or the motor was switched on or off by command.	Nothing. This is a confirmation.
Silence	Everything is normal — or the buzzer has been muted.	Check the Device Buzzer tile if you expected a sound and got none.

9.2 Sensor Node Buzzer (at the tank)

Sound	What it means	What to do
Short beep every second	Tank level has fallen below 20%.	Check that the motor can run — supply available, Tank Auto Fill on.
Long beep every second	Tank level has risen above 88%.	The motor should be stopping. If it is not, stop it from the panel and check the stop level setting.
Two short beeps	The Sensor Node has just started up and connected.	Nothing. Normal.

Silencing either buzzer. The controller's buzzer is silenced with **Stop Buzzer**; the tank buzzer with **Stop Tank Buzzer** or the red alarm banner. Both re-arm themselves — see sections 2.11 and 2.12.

9.3 On-Screen Messages

Message	Meaning and action
"Please set the clock"	Schedules cannot be accepted until the device knows the real time. Use Config Clock , or turn Remote mode on and let it sync from the internet.
Motor ON blocked	Auto Start has already started the motor on incoming municipal water. Nothing is wrong — the motor is already running.
Tank mode blocked	The motor is already running under Auto Start or Direct Control. Stop it first if you really want tank control to take over.
"no reading yet" / an old tank reading age	The Sensor Node is not reaching the controller. See section 10.
Stop level above 87% refused	Above that the probe cannot see the water. Section 4.6 explains the proper fix.
"Device Not Connected" in the header	Your phone or laptop has lost the link to the controller. Check you are still on the right Wi-Fi network.

10. Troubleshooting

The problems people actually run into, and what fixes them — in the order worth trying.

10.1 Tank Level

Symptom	Most likely cause	What to do
Tank level shows "no reading yet"	The Sensor Node is not reporting.	Check the node has power. Check its LED: lit means Wi-Fi is down. If the tank is far from the motor, fit a Wi-Fi extender (section 7).
Tank reading is minutes old	A marginal Wi-Fi link.	Move or add an extender. Confirm the node is within reach of the network.
Level sticks near the top and never reaches the stop value	The water surface is inside the probe's blind zone.	This is a mounting matter, not a fault. Lower the stop level, or remount the probe higher — section 4.6.
Level jumps around	Reflections off the tank wall, or a tilted probe.	Mount the probe level, pointing straight down, away from the wall and away from the inlet splash.
Motor stops as soon as tank mode is turned on	The tank is already above the stop level, or the node is silent so the tank is assumed full.	Check the live level and its age on the panel.

10.2 Municipal Water Detection

Symptom	Most likely cause	What to do
Motor starts when no real supply has arrived	Detection is too sensitive for your line.	Raise Flow rate to detect water , or raise Volume to confirm supply — section 5.14.
A genuine supply is missed	Detection is not sensitive enough for a weak supply.	Lower Flow rate to detect water .
Motor stops and restarts repeatedly during a supply	The cut-off rate is too close to the detection rate.	Widen the gap: lower the cut-off value.
Litre count looks wrong	The flow sensor is partly blocked or misfitted.	Have the sensor inspected. It must be fitted the right way round, in a full pipe.
Very short sessions clutter the log	They do not — very short bursts are ignored by design.	Nothing to do.

10.3 Motor and Schedules

Symptom	Most likely cause	What to do
Motor will not start from the panel	A mode conflict.	Read the message under the button, then see section 8.2.
Schedule did not run	Clock unset, or another mode had the motor.	Set the clock; check Last & Next Run to see what the device thinks is due.
Motor runs longer than the schedule duration	Another mode took over during the run — for example municipal water arrived.	Check the run-time breakdown for that day (section 5.9) to see which source is credited.
Motor restarted by itself after a power cut	Working as designed.	The motor state is restored deliberately, so an interrupted fill continues.
Run time looks far too high	Usually a leak or a tap left open.	Look at which source is climbing. "Tank auto-fill" rising means the tank is emptying faster than it should.

10.4 Connectivity, Power and Updates

Symptom	Most likely cause	What to do
Web panel will not load	Phone is on the wrong Wi-Fi network.	Reconnect to ESP32-WROOM and reload http://192.168.4.1
Header shows the device is not connected	The link dropped.	Wait a few seconds — the panel retries by itself. If it persists, reconnect to the network.
App works at home but not away	Remote / IoT mode is off.	Turn it on from the panel or the app while you are still at home, and confirm remote access works before you leave.
Repeated "Brownout" entries in Reset History	Supply voltage is dipping, usually as the motor starts.	Have an electrician check the supply and wiring at the device.
Cloud update says no newer version	You are already up to date.	Nothing to do. Check the running version under Firmware Slots .
Something looks wrong immediately after an update	—	Use Roll Back to Previous Firmware (section 5.15) and report what you saw.
Buzzer never sounds	The Device Buzzer tile is muted.	Tap the tile to un-mute. The setting survives power cuts, so it stays muted until you change it.

Before contacting support, open the web panel and note: the firmware version under **Firmware Slots**, the most recent entries in **Reset History**, and a photograph of the **Diagnostics** panel. Those three almost always identify the problem straight away.

11. Complete Feature Checklist

Everything the current BSR AutoMotor System does, in one list.

11.1 Host Controller

- Municipal water auto start on incoming flow
- Adjustable detection rate, cut-off rate and confirmation volume
- 3-minute minimum run time to prevent false stops
- Tank auto fill with anti-chatter hysteresis
- Adjustable tank start and stop levels
- Stop-level safety cap matched to the probe's real range
- Direct motor control, on and off
- Up to five daily schedules, each repeating or one-shot
- Automatic roll-forward of passed and repeating schedules
- Next-schedule preview
- Litre-accurate water quantity measurement
- Last ten municipal water sessions logged with litres and duration
- Live session marked and updated in real time
- Short-burst filtering so the log stays meaningful
- In-progress sessions checkpointed against power cuts
- 31 days of daily motor run time
- Run time split six ways by what started the motor
- Daily bar chart with per-day breakdown
- Local-midnight day boundaries
- Last-run source and time record
- Four-pattern buzzer alert system
- Buzzer mute and temporary silence
- Remote silence of the tank buzzer, with automatic re-arm
- Battery-backed real-time clock
- Automatic internet time synchronisation
- Manual clock setting from panel or app
- Ambient temperature measurement
- Always-on local Wi-Fi network
- Simultaneous router connection when remote mode is on
- Signal strength monitoring with automatic fallback
- Automatic reconnection and recovery
- Secure cloud sync for remote monitoring and control
- Built-in web control panel served to any browser
- Cloud firmware update
- Local firmware upload over Wi-Fi
- Remotely triggered firmware update
- Two firmware copies with one-tap rollback
- Automatic revert if an update fails to come up
- Permanent, power-cut-safe settings memory
- Restart cause recording, last 16 plus lifetime totals
- Device uptime and restart counters
- Data download as a plain text file
- Selective data erase that keeps your settings
- Full factory reset that keeps both firmware copies
- Physical tank-mode rocker switch
- Confirmation prompts on every destructive action

11.2 Sensor Node

- Waterproof ultrasonic level measurement
- Five-sample median filtering per reading
- Configurable tank depth and probe offset
- Invalid-echo detection and held-reading flagging
- Reporting every five seconds
- Low-level and high-level local buzzer alarms
- Remote buzzer silence honoured within one cycle
- Automatic buzzer re-arm if the controller goes quiet
- Long-range Wi-Fi configuration
- Automatic reconnection
- LED status and transmission indication

- Five-minute wireless update window after each restart

11.3 Web Control Panel

- No installation, no account, no internet required
- Phone-first single-screen motor control
- Animated live motor state
- One-tap mode tiles for auto start, tank fill and buzzer
- Live status with reading-age display
- Collapsible sections and expand-all
- Schedule editor with load and save
- Tank level and detection sensitivity settings
- Run-time chart with tap-to-inspect days
- Water session and reset history lists
- Firmware slot display and rollback
- Cloud update and local file upload with progress
- Data download, erase and factory reset
- Tank alarm banner with one-tap silence
- Audible municipal-water arrival alert in the browser
- Diagnostics log for support
- Confirmation prompts before anything irreversible

11.4 Mobile Application

- Live dashboard of every reading
- Motor on / off from anywhere
- Tank auto fill and auto start toggles
- Schedule set, view and delete
- Buzzer silence
- Clock setting
- Remote / IoT toggle
- Device firmware update
- App self-update
- Ping and water status checks
- Works on home Wi-Fi and on mobile data worldwide

12. Quick Start — Five Steps

From an installed system to full automation, in about ten minutes.

Step 1 — Install the Hardware

- Mount the **Host Controller** near the motor or the electrical panel and connect the motor through it.
- Fit the **flow sensor** in the pipe between the municipal supply and the motor, the right way round.
- Mount the **Sensor Node** above the tank, pointing straight down, ideally at least 25 cm above the full mark (section 4.6).
- If the tank is far from the motor, place a **Wi-Fi extender** midway between them.
- Power everything up. The controller gives one long beep; the Sensor Node gives two short ones once it has connected.

Step 2 — Connect Your Phone

- In Wi-Fi settings, join the network named **ESP32-WROOM** using the password supplied with your installation.
- Open a browser and go to **http://192.168.4.1**
- Tap the notice at the top once, to allow the page to play the water-arrival alert sound.
- Confirm the header shows **Device Connected**.

Step 3 — Set the Clock

- Open **Controls** and tap **Config Clock**.
- Enter the current date and time and confirm.
- Alternatively, turn on Remote / IoT mode (step 5) and let the device sync from the internet automatically.

Schedules will not be accepted until this is done.

Step 4 — Choose Your Mode

- For most homes, leave **Municipal Water Auto Start** ON and turn **Tank Auto Fill** ON as well.
- Open **Tank Levels**, tap **Get From Device**, and adjust the start and stop levels if the defaults of 50% and 87% do not suit your tank.
- If your supply arrives at a fixed time, add a repeating schedule under **Schedules** and tap **Save All**.

Step 5 — Enable Internet (Optional but Recommended)

- Open **Remote / IoT Mode** and switch it on. The panel may drop for a few seconds while the device changes Wi-Fi mode — this is normal.
- Install the mobile app and sign in.
- **Test remote access before you need it:** turn Wi-Fi off on your phone, and confirm the app still reads your device over mobile data.

Step 6 — After a Week, Look at the Data

- **Municipal Water sessions** tells you when your supply really arrives and how much it delivers.
- **Motor Run Time** over 7 days tells you what is actually driving your motor.
- Use both to fine-tune your schedules and your tank levels. That is where a good installation becomes a very good one.

13. Specifications, Care & Safety

13.1 System Summary

Item	Detail
Tank level reporting interval	Every 5 seconds
Tank reading method	Ultrasonic, median of 5 samples per cycle
Default tank start / stop levels	50% / 87% (both adjustable)
Maximum permitted stop level	87%
Tank alarm band at the Sensor Node	Below 20%, above 88%
Sensor Node timeout safety cut-off	2 minutes without a reading
Default water detection rate	0.5 litres/minute (adjustable)
Default cut-off rate	0.4 litres/minute (adjustable)
Default confirmation volume	100 ml within a 2-minute window (adjustable)
Minimum motor run time on auto start	3 minutes
Schedules	Up to 5, each repeating daily or one-shot
Water session history	Last 10 sessions
Motor run-time history	31 days, split by 6 trigger sources
Restart history	Last 16 restarts, plus lifetime totals per cause
Firmware copies held	2 (running plus one step back)
Temperature measurement	Accurate to about 0.25 °C
Local network name	ESP32-WROOM
Local control panel address	http://192.168.4.1
Typical local Wi-Fi range	10–20 m indoors, extendable with a repeater
Mobile app platform	Android

13.2 Routine Care

- **Every month** — glance at Motor Run Time. A change in the pattern is usually the first sign of a leak or a failing float.
- **Every three months** — check the ultrasonic probe is still clean, level and unobstructed. Dust, cobwebs and nests all affect readings.
- **Every three months** — check the Reset History for brownouts.
- **Every six months** — have the flow sensor inspected for scale or debris.
- **Before travelling** — confirm remote access works over mobile data while you are still at home.
- **When a firmware update is offered** — install it. Updates carry both fixes and new features, and rollback is always available if anything looks wrong.

13.3 Safety

Electrical work must be done by a qualified electrician. The Host Controller switches mains-powered motor equipment. Installation, wiring and any later change to the motor circuit are jobs for a licensed professional, in line with local regulations.

- **Always isolate the supply** at the mains before opening any enclosure or working on the motor.
- **The motor can start without warning.** That is the entire point of the product. Never assume a stopped motor will stay stopped — isolate the supply before any mechanical work on the pump, the pipework or the tank.
- **Keep the enclosure closed and dry.** The Host Controller is not designed for direct exposure to water or weather.
- **Protect against dry running.** This system controls when the motor runs, but it is not a substitute for the dry-run protection your pump itself should have.
- **Mount the Sensor Node securely.** A probe that falls into the tank stops measuring and contaminates your water.
- **Do not modify the device** or install firmware from any source other than BSR.
- **Supervise tank access.** Open water tanks are a drowning hazard, particularly to children.

13.4 What the System Does Not Do

Stated plainly, so expectations are right from day one:

- It does not measure water quality — only quantity and level.
- It cannot measure the very top of the tank unless the probe is mounted high enough (section 4.6).
- It cannot create water pressure. If your supply is weak, the system will detect and report that, but it cannot fix it.
- It does not replace your pump's own overload or dry-run protection.

- It will not run the motor during a mains power cut. When power returns, it restores exactly the state it was in.

14. Glossary

Term	Meaning
Auto Start / AutoFill	The mode that starts the motor automatically when municipal water is detected in the supply pipe.
Tank Auto Fill	The mode that starts and stops the motor based on how full the tank is.
Direct Motor Control	Turning the motor on or off yourself, overriding the automatic modes.
Hysteresis / dead band	The deliberate gap between the level that starts the motor and the level that stops it, which prevents rapid on-off switching.
Blind zone	The minimum distance an ultrasonic probe can measure. Anything closer than that returns no usable reading.
Median filtering	Taking several readings and using the middle value, which discards occasional wild results.
Flow rate	How fast water is moving through the pipe, in litres per minute.
Session	One continuous municipal water supply, from the moment it is detected to the moment it stops.
Trigger source	What caused the motor to start: mobile app, web page, schedule, municipal water, tank auto-fill or the physical switch.
Firmware	The software running inside the device itself.
Firmware slot	One of the two places in the device where a firmware copy is kept.
Rollback	Restoring the previous firmware copy.
OTA	"Over the air" — updating the device wirelessly, with no cable and no physical access.
Brownout	A dip in supply voltage severe enough to restart the device.
Uptime	How long the device has been running since its last restart.
Local Wi-Fi network	The network the Host Controller creates itself. It exists whether or not you have internet.
Remote / IoT mode	The setting that connects the device to your router and the cloud, enabling control from outside your home.
Wi-Fi extender / repeater	A device that rebroadcasts a Wi-Fi network to cover a greater distance.

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